

COURSE TITLE : PLASTIC PRODUCTS
COURSE CODE : 6091
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75/6
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Basic factors of plastic product manufacture	19
2	Injection and Blow moulded plastic products	18
3	Thermoforming , Extrusion and Calendering	19
4	Miscellaneous Plastic products, Finishing and decoration of plastic products	19
TOTAL		75

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To understand the basic factors of plastic product manufacture
2	2	To comprehend the plastic product manufacture by Injection moulding and blow moulding
3	3	To know the plastic products produced by thermoforming , extrusion and calendering
4	4	To understand the cast plastic products, expanded products ,powder coated products, compression and Transfer moulded products
	5	To understand the different methods of finishing and decoration of plastic products

SPECIFIC OUTCOME

MODULE I BASIC FACTORS OF PLASTIC PRODUCT MANUFACTURE

1.1.0 To understand the basic factors of plastic product manufacture

- 1.1.1 List major plastic products based on commodity and engineering plastics
- 1.1.2 Classify the plastics products according to process of manufacture

- 1.1.3 Explain different factors (functional and aesthetic) to be considered while design a plastic products
- 1.1.4 Describe the various steps involved in the manufacture of plastics products
- 1.1.5 List the types of ingredients used in plastic products
- 1.1.6 State the function , properties and applications of fillers ,stabilisers ,curing agents , plasticizers, colourants ,flow promoturs and antidegradants in plastic products
- 1.1.7 Explain the rheological behaviour of plastics in product manufacture
- 1.1.8 Explain the stress concentration and frozen- in stress in product manufacture
- 1.1.9 Mention the environmental impact of plastics proliferation

MODULE II INJECTION AND BLOW MOULDED PLASTIC PRODUCTS

2.2.0 To comprehend the plastic product manufacture by Injection moulding and blow moulding.

- 2.2.1 Identify different injection moulded products
- 2.2.2 Mention the characteristics of plastics required for injection moulding
- 2.2.3 State the properties of injection moulding grades of HDPE, PS, Nylon , PVC and Polyester
- 2.2.4 Describe the requirements of injection moulds and production methods of bucket, mugs, bottle cap, pen, scale and chairs .
- 2.2.5 Explain the manufacture of PVC Chappal with typical formulation.
- 2.2.6 Expain the method of production and raw materials for nylon gears
- 2.2.7 Identify different blow moulded products
- 2.2.8 Mention the characteristics of plastics required for blow moulding
- 2.2.9 Describe the process and raw materials for the manufacture of PVC Bottles and PET bottles

MODULE III THERMOFORMING, EXTRUSION AND CALENDERING

3. 3.0 To know the plastic products produced by thermoforming , extrusion and calendering

- 3.3.1 Identify various plastic products produced by thermoforming, extrusion and calendaring
- 3.3.2 Describe the production process of disposable plastic glass and egg trays from poly olefins
- 3.3.3 Explain the design factors such as standard dimension ratio, water hammer and stresses (external and internal)
- 3.3.4 State the equation and simple calculations for SDR and wall thickness for different hoop stress
- 3.3.5 Explain the manufacture of PVC Pipe (rigid and flexible) with typical formulation
- 3.3.6 Draw and explain line flow chart for PVC and PE pipe extrusion
- 3.3.7 Explain the differences in the design of dies of PVC and PE pipe with sketches
- 3.3.8 Explain the manufacture of blown films .
- 3.3.9 Explain the manufacture of PVC rexin sheets and vinyl flooring with typical formulations.

MODULE IV MISCELLANEOUS PLASTIC PRODUCTS, FINISHING AND DECORATION OF PLASTIC PRODUCTS

4.4.0 To understand the cast plastic products, expanded products , powder coated products, compression and Transfer moulded products

- 4.4.1. Identify plastic products produced by Casting ,Compression moulding ,Transfer moulding, foaming and Coating .
- 4.4.2. Explain the manufacturing of PMMA sheets
- 4.4.3. Describe Bakelite and Meloware moulding

- 4.4.4. Explain the manufacture of inserted plastic products
- 4.4.5. Explain the manufacture of rigid and flexible PU foam .
- 4.4.6. Describe the manufacture of powder coated plastic products

4.5.0 To understand the different methods of finishing and decoration of plastic products

- 4.5.1. State the purpose of finishing and machining operations of plastic products
- 4.5.2 Explain the finishing operations such as tumbling , grinding and sanding ,trimming ,joining and assembling and buffing
- 4.5.3. List the various finishing operations of plastic products
- 4.5.4. Describe the processes –Printing , Electro plating ,Painting ,Coating , Lazer marking, Vacuum metalizing and labeling of plastic products

CONTENT DETAILS

MODULE-I

Important Plastics -General idea about plastic product design. Plastic product based on commodity and engineering plastic and thermo set plastics .Classify plastic product –functional and aesthetic design – steps involved in plastic product design –ingredients used in plastic products . fillers stabilizers curing agents plasticizers colorants flow promo tents anti degradedness. Rheological behavior of plastics stress concentration frozen in stress . environmental impact of plastic product proliferation .

MODULE-II

Injection and blow moulded plastic products . General features of injection moulded products-based on HDPE, PP, and other poly olefins bucket, mug, bottle caps, pen, scale ,chair ect. s .Products based on PVC, PS, nylon polyester . manufacture of PVC chapel with formulation. Production of nylon gears . Different blow moulded products PVC Bottles ,Pet Bottles , Characteristics of plastic required for bow moulding .

MODULE-III

Thermoforming Extrusion and calendared products- various plastic products produced by thermoforming –extrusion and calendaring. Production of disposable glass –egg trays from poly olefins. Production of PVC Pipe (flexible and rigid with typical formulation)–design factors of PVC Pipes such as SDR, Water hammer and stress State the equation and simple problems for SDR .Production of PE Pipe Difference in dies of PVC and PE Pipe with sketches . Line Flow diagram of PVC and PE pipe extrusion. Manufacture of blown film extrusion . Manufacture of PVC Rixin – vinyl flooring –Typical formulations.

MODULE-IV

Miscellaneous plastic products , finishing and decoration of plastic products -by casting – compression moulding –foaming coating manufacture of PMMA sheets – Bakelite and meloware moulded products –

inserted plastic products – rigid and flexible PU foam – Powder coated plastic products ,Finishing and machining of plastic products –tumbling –grinding –sanding –trimming joining –assembling and buffing decoration of plastic products –printing –electro plating –painting –coating –laser marking –vacuum metalizing and labeling of plastic products

REFERENCES

1. Plastic Product Design Engineering Hand book – Sydney Levy, J Harry DuBois (Springer US,1984)
2. Blow moulding -David A Jenners
3. Injection moulding -Irvin Ruhin
4. Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing Co. 1986)
5. Plastics Engineering Handbook Of The Society Of The Plastics Industry – M. Berins, (Publisher - Springer Science & Business Media, 1991)